

# Notes: Kaniel, Liu, Saar & Titman (JF, 2012)

## Individual Investor Trading and Return Patterns around Earnings Announcements

### Contents

|          |                                                                                                                                                                              |           |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Big picture</b>                                                                                                                                                           | <b>3</b>  |
| <b>2</b> | <b>Incremental contribution of the paper</b>                                                                                                                                 | <b>3</b>  |
| <b>3</b> | <b>Data and measurement</b>                                                                                                                                                  | <b>4</b>  |
| 3.1      | Sample construction . . . . .                                                                                                                                                | 4         |
| 3.2      | Individual investor trading measure . . . . .                                                                                                                                | 4         |
| 3.3      | Earnings surprise measure . . . . .                                                                                                                                          | 4         |
| 3.4      | Return outcomes . . . . .                                                                                                                                                    | 5         |
| <b>4</b> | <b>Models and empirical specifications</b>                                                                                                                                   | <b>5</b>  |
| 4.1      | Portfolio sorts on pre-event individual trading . . . . .                                                                                                                    | 5         |
| 4.2      | Controls for earnings surprise and past returns . . . . .                                                                                                                    | 5         |
| 4.3      | Liquidity provision decomposition . . . . .                                                                                                                                  | 5         |
| 4.4      | Post-event trading regression . . . . .                                                                                                                                      | 6         |
| <b>5</b> | <b>Identification and empirical design</b>                                                                                                                                   | <b>6</b>  |
| 5.1      | What identifies informed individual trading? . . . . .                                                                                                                       | 6         |
| 5.2      | Key threats and responses . . . . .                                                                                                                                          | 6         |
| 5.3      | Why size heterogeneity matters . . . . .                                                                                                                                     | 6         |
| 5.4      | Dividend announcement placebo/extension . . . . .                                                                                                                            | 7         |
| <b>6</b> | <b>Tables and figures</b>                                                                                                                                                    | <b>7</b>  |
| 6.1      | Table I: Summary Statistics; Table II: Net Individual Trading around Earnings Announcements                                                                                  | 7         |
| 6.2      | Table III: Relation between Pre-event Individual Trading and Return Predictability; Table IV: Controlling for Past Returns and Earnings Surprises . . . . .                  | 8         |
| 6.3      | Table V: Regressions Relating Pre-event Trading of Individuals to Future Abnormal Returns                                                                                    | 9         |
| 6.4      | Table VI: Decomposition of the Abnormal Returns following Individual Trading . . . . .                                                                                       | 9         |
| 6.5      | Table VII: Decomposition of the Abnormal Return with Risk Adjustment . . . . .                                                                                               | 10        |
| 6.6      | Table VIII: Analysis of Return Predictability around Dividend Announcements . . . . .                                                                                        | 11        |
| 6.7      | Figure 1: Investor Trading Conditional on Earnings Surprise; Figure 2: Investor Trading Conditional on Earnings Surprise and Pre-event Returns . . . . .                     | 12        |
| 6.8      | Table IX: Post-event Individual Trading Conditional on Earnings Surprises and Pre-event Returns; Table X: Regressions Explaining Post-event Net Individual Trading . . . . . | 12        |
| <b>7</b> | <b>Short summary</b>                                                                                                                                                         | <b>13</b> |

|          |                                                                                 |           |
|----------|---------------------------------------------------------------------------------|-----------|
| <b>8</b> | <b>Conclusion</b>                                                               | <b>13</b> |
| 8.1      | Core conclusion . . . . .                                                       | 13        |
| 8.2      | Economic intuition . . . . .                                                    | 13        |
| 8.3      | Why the distinction between pre-event and post-event behavior matters . . . . . | 14        |
| 8.4      | Implications for market efficiency . . . . .                                    | 14        |
| 8.5      | Limitations and caveats . . . . .                                               | 14        |
| 8.6      | Takeaway . . . . .                                                              | 14        |

# 1 Big picture

- **Question.** Do individual investors trade on information around earnings announcements, or are their trades mainly uninformed liquidity provision/noise trading?
- **Core finding.** Aggregated individual investor buying before earnings announcements predicts positive abnormal returns on and after the announcement; aggregated selling predicts negative abnormal returns.
- **Magnitude.** Stocks intensely bought by individuals in the 10 days before an earnings announcement earn materially higher subsequent abnormal returns than stocks intensely sold, with a large spread around the announcement and over the following months.
- **Mechanism.** The paper decomposes return predictability into liquidity-provision compensation and an information component. About half of the return pattern is attributed to private information or interpretive skill.
- **Post-announcement behavior.** After earnings announcements, individuals trade in a return-contrarian manner and also in a news-contrarian manner, buying after bad earnings surprises and selling after good surprises.
- **Asset-pricing implication.** Individual trading is not simply noise. Aggregated retail order flow contains information relevant for future returns, especially for smaller stocks.
- **Market-efficiency implication.** If individuals sell after good news and buy after bad news, their post-event trading can slow the adjustment of prices to earnings news and contribute to post-earnings-announcement drift.

# 2 Incremental contribution of the paper

This paper makes several incremental contributions relative to prior research on individual investors and earnings announcements.

- **Direct measurement of individual trades.** Many earlier papers infer individual trading from small trades. This paper uses NYSE data that directly identify aggregate individual-investor buying and selling, avoiding the trade-size proxy problem.
- **Large sample of announcements.** The analysis covers 1,821 NYSE stocks and 17,564 earnings announcements from 2000–2003, providing much more statistical power than older datasets such as TORQ.
- **Event-centered information test.** Rather than asking whether retail trading predicts returns unconditionally, the paper focuses on earnings announcements, when private information and information-processing skill should be most valuable.
- **Liquidity versus information decomposition.** The paper goes beyond documenting return predictability by decomposing subsequent returns into a liquidity-provision component and an information component.
- **Post-news trading behavior.** The paper separates return-contrarian trading from news-contrarian trading after earnings announcements, showing that individuals trade against the earnings surprise itself.
- **Reconciliation with behavioral and market-making views.** The results show that individuals can look informed before announcements while also behaving contrarian after public news. This richer pattern helps reconcile informed-retail and behavioral-retail interpretations.

**Interpretation:** The incremental contribution is not simply that individual trading predicts returns. The paper shows when this predictability is strongest, how much appears informational rather than liquidity-related, and how post-news individual trading can slow price discovery.

### 3 Data and measurement

#### 3.1 Sample construction

The sample consists of all common domestic stocks traded on the NYSE between January 2000 and December 2003 that can be matched to CRSP, I/B/E/S, and the NYSE individual-investor trading data.

- Earnings announcement dates are obtained from I/B/E/S and COMPUSTAT.
- The final sample contains 1,821 stocks and 17,564 earnings announcement events.
- For each event, the paper requires 60 days of data before and after the announcement.
- Stocks are sorted into market-capitalization groups: small, mid-cap, and large.

**Interpretation:** The setting is designed to study information-sensitive trading around scheduled corporate news. Requiring data around the event makes the return windows and order-flow windows comparable across stocks.

#### 3.2 Individual investor trading measure

The paper observes daily aggregated buying and selling volume by individuals from NYSE data. For each stock-day, individual net trading is constructed as a normalized imbalance:

$$Imb_{it} = \frac{BuyVol_{it}^{ind} - SellVol_{it}^{ind}}{AvgDollarVol_i}.$$

The paper then subtracts the stock’s average daily individual imbalance over the full sample period:

$$IndNT_{it} = Imb_{it} - \overline{Imb}_i.$$

For event windows, the daily measures are cumulated:

$$IndNT_{i,[a,b]} = \sum_{t=a}^b IndNT_{it}.$$

**Interpretation:** Normalizing by average dollar volume makes individual trading comparable across stocks of different size. Subtracting the stock-specific average removes persistent retail buying or selling tendencies.

#### 3.3 Earnings surprise measure

The earnings surprise measure compares announced earnings to the mean analyst forecast from I/B/E/S. Stocks are sorted each quarter into five earnings-surprise quintiles:

- *ES1*: most negative earnings surprise;
- *ES5*: most positive earnings surprise.

**Interpretation:** Sorting by earnings surprise allows the paper to ask whether individual post-announcement trading responds to the direction of public news, not only to price movements.

### 3.4 Return outcomes

The paper uses cumulative abnormal returns over multiple windows around earnings announcements:

$$CAR_{i,[a,b]} = \sum_{t=a}^b AR_{it}.$$

The most important windows include:

- $[-10, -1]$ : pre-announcement return window;
- $[0, 1]$ : two-day announcement window;
- $[0, 61]$ : announcement plus post-announcement drift window;
- $[2, 61]$ : post-announcement drift excluding the event window.

**Interpretation:** The short event window captures immediate price discovery, while the longer windows capture delayed price adjustment and post-earnings-announcement drift.

## 4 Models and empirical specifications

### 4.1 Portfolio sorts on pre-event individual trading

Each quarter, stocks with earnings announcements are sorted into quintiles using pre-event net individual trading:

$$IndNT_{[-10,-1]} \quad \text{or} \quad IndNT_{[-60,-1]}.$$

The paper compares subsequent abnormal returns across the low and high individual-trading quintiles.

**Interpretation:** This is the simplest test of whether individuals are informed before earnings announcements. If pre-event buying reflects information, high-*IndNT* stocks should earn higher future returns than low-*IndNT* stocks.

### 4.2 Controls for earnings surprise and past returns

A regression version relates future abnormal returns to pre-event individual trading, earnings surprises, and past abnormal returns:

$$CAR_{i,[0,61]} = \alpha + \beta_1 IndNT_{i,[-10,-1]} + \beta_2 IndNT_{i,[-60,-1]} + \gamma ES_i + \delta CAR_{i,pre} + \varepsilon_i.$$

**Interpretation:** The coefficient on *IndNT* asks whether individual trading predicts returns beyond information already contained in analyst earnings surprises and past stock returns.

### 4.3 Liquidity provision decomposition

The paper decomposes returns following individual trading into components attributed to liquidity provision and information. A simplified version of the logic is:

$$CAR_{i,t+1} = a_t + b_t IndNT_{i,t} + c_t CAR_{i,pre} + e_{i,t+1},$$

where the return component associated with the price of liquidity provision is separated from the component that remains after accounting for liquidity demand and reversal.

**Interpretation:** If individuals provide liquidity to impatient institutional traders, they should earn compensation even without private information. The decomposition is designed to ask how much of the predictability is liquidity compensation and how much looks like information.

## 4.4 Post-event trading regression

To explain individual trading after the announcement, the paper regresses post-event individual net trading on earnings-surprise dummies, pre-event returns, and event-window returns:

$$IndNT_{i,post} = \alpha + \gamma ES_i + \delta CAR_{i,[-60,-1]} + \phi CAR_{i,[0,1]} + u_i.$$

**Interpretation:** A negative relationship with the earnings surprise indicates news-contrarian trading. A negative relationship with pre-event returns indicates return-contrarian trading. The paper finds both.

# 5 Identification and empirical design

## 5.1 What identifies informed individual trading?

The core empirical design is event-based. Earnings announcements are predictable dates when firm information is released. If individuals trade before these dates and their trades predict announcement and post-announcement abnormal returns, this suggests that their trades contain information.

**Interpretation:** The identifying logic is not a randomized experiment. It is a timing and prediction design: pre-announcement retail order flow is measured before public news is released, and the outcome is subsequent abnormal return behavior.

## 5.2 Key threats and responses

- **Liquidity provision.** Individuals may earn returns by taking the other side of temporary liquidity demand, not by having information. The paper directly addresses this through its return-decomposition methodology.
- **Past-return reversal.** Individuals may be buying losers and selling winners. The paper controls for pre-event abnormal returns and sorts jointly on pre-event returns and trading.
- **Public earnings news.** Return predictability may simply proxy for analyst earnings surprises. Regressions include earnings-surprise controls and show that individual trading adds predictive power.
- **Small-trade measurement error.** The paper avoids the small-trade proxy problem by using direct NYSE individual trading data.
- **Clustering of announcements.** The paper uses Fuller-Battese random-effect methods to account for event clustering in calendar time.

## 5.3 Why size heterogeneity matters

Information-based trading should be stronger among smaller stocks because these firms have less analyst and institutional coverage. The paper finds stronger information components for small firms.

**Interpretation:** The size pattern is an important validation check. Liquidity provision can operate in any size group, but private retail information is more plausible where formal information production is weaker.

## 5.4 Dividend announcement placebo/extension

The paper also studies dividend announcements. Dividend announcements are less information-rich than earnings announcements, but they provide a related corporate event. The results around dividends are weaker but still consistent with some information in individual trading.

**Interpretation:** Dividend tests help distinguish a pure earnings-specific anomaly from broader event-related individual trading ability.

## 6 Tables and figures

### 6.1 Table I: Summary Statistics; Table II: Net Individual Trading around Earnings Announcements

**Read:**

- Table I shows the event sample: 1,821 stocks and 17,564 earnings announcements, with market capitalization, price, turnover, volume, and return volatility by size group.
- Table II shows that individuals tend to buy before earnings announcements and sell during and after the announcement.
- The sign reversal is important: individuals accumulate before the information event and reduce positions around or after the public announcement.

**Economic message:** The basic trading pattern already suggests that individuals do not simply chase attention at the announcement. They buy before the announcement and sell when news arrives, consistent with advance positioning and later unwinding.

**Interpretation:** Table II is the empirical foundation for the paper. It defines the pre-event trading variables later used to predict returns and shows that individual trading is systematically timed around earnings events.

**Table I**  
**Summary Statistics**

The sample of stocks for the study consists of all common domestic stocks traded on the NYSE at any time between January 1, 2000 and December 31, 2003 with records in the CRSP database. We use ticker symbol and CUSIP to match the stocks to a data set containing daily aggregated buying and selling volume of individuals provided to us by the NYSE. We then use I/B/E/S and COMPUSTAT to identify the dates on which our sample stocks had earnings announcements. We impose two restrictions on the sample. First, we require 60 days of data prior to and after the announcements. Second, we require that there be an observation in the I/B/E/S database for the mean analyst forecast in the month prior to the earnings announcement (and also the actual earnings number). Our screens result in a final sample of 1,821 stocks with 17,564 earnings announcement events. In Panel A, we provide summary statistics from the CRSP database. For each stock we compute the following time-series measures: AvgCap is the average monthly market capitalization over the sample period; AvgPrc is the average daily closing price; AvgTurn is the average weekly turnover (number of shares traded divided by the number of shares outstanding); AvgVol is the average weekly dollar volume; and StdRet is the standard deviation of weekly returns. We then sort the stocks by market capitalization into 10 deciles, and form three size groups: small stocks (deciles 1, 2, 3, and 4), mid-cap stocks (deciles 5, 6, and 7), and large stocks (deciles 8, 9, and 10). The cross-sectional mean and median of these measures are presented for the entire sample and separately for the three size groups. In Panel B, we provide the number of earnings announcement events used in the analysis for each month during the sample period.

| Panel A: Summary Statistics of Sample Stocks (from CRSP)      |        |                           |                   |                  |                           |                 |       |       |     |       |     |     |
|---------------------------------------------------------------|--------|---------------------------|-------------------|------------------|---------------------------|-----------------|-------|-------|-----|-------|-----|-----|
|                                                               |        | AvgCap<br>(in million \$) | AvgPrc<br>(in \$) | AvgTurn<br>(in%) | AvgVol<br>(in million \$) | StdRet<br>(in%) |       |       |     |       |     |     |
| All stocks                                                    | Mean   | 5,783.5                   | 64.16             | 2.67             | 125.00                    | 7.26            |       |       |     |       |     |     |
|                                                               | Median | 1,049.8                   | 22.87             | 2.19             | 27.06                     | 6.11            |       |       |     |       |     |     |
| Small stocks                                                  | Mean   | 354.5                     | 15.49             | 2.65             | 11.34                     | 8.84            |       |       |     |       |     |     |
|                                                               | Median | 353.2                     | 12.40             | 1.83             | 5.86                      | 7.36            |       |       |     |       |     |     |
| Mid-Cap stocks                                                | Mean   | 1,367.5                   | 27.28             | 3.29             | 45.74                     | 6.76            |       |       |     |       |     |     |
|                                                               | Median | 1,279.6                   | 24.37             | 2.62             | 34.15                     | 6.01            |       |       |     |       |     |     |
| Large stocks                                                  | Mean   | 14,652.0                  | 140.38            | 3.25             | 321.40                    | 6.07            |       |       |     |       |     |     |
|                                                               | Median | 5,314.5                   | 37.59             | 2.61             | 170.62                    | 5.32            |       |       |     |       |     |     |
| Panel B: Number of Earnings Announcement Events in Our Sample |        |                           |                   |                  |                           |                 |       |       |     |       |     |     |
|                                                               | Jan    | Feb                       | Mar               | Apr              | May                       | Jun             | Jul   | Aug   | Sep | Oct   | Nov | Dec |
| 2000                                                          | 0      | 0                         | 17                | 949              | 343                       | 84              | 929   | 345   | 90  | 786   | 288 | 86  |
| 2001                                                          | 638    | 488                       | 160               | 852              | 283                       | 82              | 829   | 338   | 71  | 866   | 289 | 78  |
| 2002                                                          | 626    | 456                       | 120               | 843              | 304                       | 73              | 879   | 282   | 78  | 903   | 272 | 87  |
| 2003                                                          | 589    | 510                       | 148               | 851              | 318                       | 75              | 879   | 290   | 80  | 10    | 0   | 0   |
| All years                                                     | 1,853  | 1,454                     | 445               | 3,495            | 1,248                     | 314             | 3,516 | 1,255 | 319 | 2,565 | 849 | 251 |

(a) Table I: Summary Statistics. Sample composition and firm characteristics by size group.

**Table II**  
**Net Individual Trading around Earnings Announcements**

This table presents net individual trading around earnings announcements. We construct the net individual trading measure by first computing an imbalance measure, that is, by subtracting the daily value of the shares sold by individuals from the value of shares bought and dividing by the average daily dollar volume (from CRSP) in the calendar year. We then subtract from the imbalance measure the daily average of individual imbalances over the sample period to get the net individual trading measure, and compute for each stock the cumulative net individual trading measure over certain periods before, during, and after the announcement. Since each week contains multiple earnings announcements, we implement the Fuller-Battese (1974) methodology to correct for clustering. For each quintile, we model the net individual trading measure using a one-way random effect framework in which there is a weekly effect (for [-5, -1], [0, 1], and [2, 6]), a monthly effect (for [-20, -1], [-10, -1], [2, 11], and [2, 21]), or a quarterly effect (for [-60, -1] and [2, 61]). We report the estimated mean with clustering-corrected *t*-statistics (in parentheses, testing the hypothesis of zero net individual trading). We use \*\* to indicate significance at the 1% level and \* to indicate significance at the 5% level (both against a two-sided alternative).

|                |         | Time Periods |           |           |          |          |          |          |         |         |  |
|----------------|---------|--------------|-----------|-----------|----------|----------|----------|----------|---------|---------|--|
|                |         | [-60, -1]    | [-20, -1] | [-10, -1] | [-5, -1] | [0, 1]   | [2, 6]   | [2, 11]  | [2, 21] | [2, 61] |  |
| All stocks     | Mean    | 0.061        | 0.032     | 0.022*    | 0.017**  | -0.005*  | -0.015** | -0.016   | -0.019  | 0.048   |  |
|                | t-stat. | (0.62)       | (1.64)    | (2.30)    | (4.77)   | (-2.15)  | (-3.39)  | (-1.30)  | (-0.85) | (0.48)  |  |
| Small stocks   | Mean    | 0.116        | 0.053     | 0.034     | 0.030**  | -0.008   | -0.017   | -0.012   | -0.005  | 0.057   |  |
|                | t-stat. | (0.70)       | (1.54)    | (1.88)    | (4.24)   | (-1.66)  | (-1.84)  | (-0.50)  | (-0.11) | (0.36)  |  |
| Mid-cap stocks | Mean    | 0.000        | 0.029     | 0.021*    | 0.004    | -0.001   | -0.018** | -0.027** | -0.039* | 0.003   |  |
|                | t-stat. | (0.00)       | (1.46)    | (2.24)    | (0.38)   | (-0.40)  | (-4.07)  | (-2.71)  | (-2.05) | (0.04)  |  |
| Large stocks   | Mean    | -0.016       | 0.010     | 0.008     | 0.005**  | -0.003** | -0.008** | -0.009   | -0.013  | 0.005   |  |
|                | t-stat. | (-0.37)      | (0.98)    | (1.57)    | (2.62)   | (-2.90)  | (-4.30)  | (-1.95)  | (-1.43) | (0.11)  |  |

(b) Table II: Net Individual Trading around Earnings Announcements. Retail buying before earnings and selling around/after the event.

## 6.2 Table III: Relation between Pre-event Individual Trading and Return Predictability; Table IV: Controlling for Past Returns and Earnings Surprises

Read:

- Table III sorts events by pre-event individual trading and shows that stocks bought most intensely by individuals earn higher future abnormal returns than stocks sold most intensely.
- The spread between high and low retail-trading quintiles is economically large in the announcement window and remains meaningful over longer post-event windows.
- Table IV shows that this predictive relation is not simply a disguised past-return reversal or a mechanical response to analyst earnings surprises.

**Economic message:** Pre-event retail order flow contains information about future announcement returns. The table pair is the central evidence that individual trading has forecasting content around earnings announcements.

**Interpretation:** Table III documents the raw predictive relation. Table IV is the robustness step: by conditioning on past returns and earnings surprises, it shows that the retail-trading signal contains incremental information.



**Table III**  
**Predicting Returns Using Net Individual Trading before the Announcements**

This table presents analysis of market-adjusted returns on and after earnings announcements conditional on different levels of net individual trading before the event. We construct the net individual trading measure by first computing an imbalance measure (see Table II). We then subtract from the imbalance measure the daily average of individual imbalances over the sample period to get the net individual trading measure, and compute for each stock the cumulative net individual trading measure over the 10 days before the announcement. In Panel A, we sort all stocks each quarter into quintiles according to net individual trading in the 10 trading days prior to the announcement ( $IndNT_{[-10, -1]}$ ) (quintile 1 contains the stocks that individuals sold the most and quintile 5 contains the stocks that individuals bought the most). We then compute for each stock the cumulative market-adjusted return over certain periods. Since each week contains multiple earnings announcements, we implement the Fuller-Battese (1974) methodology to correct for clustering. For each quintile, we model the cumulative abnormal return using a one-way random effect framework in which there is a weekly effect (for [0, 1] and [2, 6]), a monthly effect (for [2, 11] and [2, 21]), or a quarterly effect (for [2, 61] and [0, 61]). We report the estimated means with clustering-corrected  $t$ -statistics (in parentheses, testing the hypothesis of zero cumulative abnormal return). In Panel B, we separately sort large, mid-cap, and small stocks into quintiles according to net individual trading before the event, and report just the row "Difference between Q5 and Q1" for each of these size groups. We use \*\* to indicate significance at the 1% level and \* to indicate significance at the 5% level (both against a two-sided alternative).

| Panel A: Predicting Returns with Pre-Event Net Individual Trading |            |              |           |           |           |           |           |
|-------------------------------------------------------------------|------------|--------------|-----------|-----------|-----------|-----------|-----------|
| $IndNT_{[-10, -1]}$                                               |            | Time Periods |           |           |           |           |           |
|                                                                   |            | [0, 1]       | [2, 6]    | [2, 11]   | [2, 21]   | [2, 61]   | [0, 61]   |
| Q1<br>(Selling)                                                   | Mean       | -0.0066**    | -0.0041** | -0.0045** | -0.0096** | -0.0281** | -0.0338** |
|                                                                   | $t$ -stat. | (-4.74)      | (-3.82)   | (-2.99)   | (-4.20)   | (-5.76)   | (-5.91)   |
| Q2                                                                | Mean       | 0.0001       | 0.0005    | 0.0007    | -0.0016   | -0.0208** | -0.0198** |
|                                                                   | $t$ -stat. | (0.06)       | (0.46)    | (0.33)    | (-0.59)   | (-3.39)   | (-3.34)   |
| Q3                                                                | Mean       | 0.0037**     | 0.0042**  | 0.0056**  | 0.0087**  | -0.0012   | 0.0030    |
|                                                                   | $t$ -stat. | (2.57)       | (3.37)    | (2.69)    | (2.76)    | (-0.21)   | (0.57)    |
| Q4                                                                | Mean       | 0.0085**     | 0.0074**  | 0.0104**  | 0.0140**  | 0.0102**  | 0.0191**  |
|                                                                   | $t$ -stat. | (6.24)       | (5.89)    | (4.99)    | (4.82)    | (2.70)    | (4.80)    |
| Q5<br>(Buying)                                                    | Mean       | 0.0078**     | 0.0031*   | 0.0057**  | 0.0096**  | 0.0139    | 0.0215**  |
|                                                                   | $t$ -stat. | (4.44)       | (2.28)    | (2.97)    | (3.19)    | (1.91)    | (2.88)    |
| Diff. bet.                                                        | Mean       | 0.0147**     | 0.0072**  | 0.0100**  | 0.0187**  | 0.0413**  | 0.0545**  |
| Q5 and Q1                                                         | $t$ -stat. | (7.48)       | (4.16)    | (4.31)    | (6.04)    | (7.80)    | (9.53)    |
| Panel B: Return Predictability by Market Capitalization Groups    |            |              |           |           |           |           |           |
| $IndNT_{[-10, -1]}$                                               |            | Time Periods |           |           |           |           |           |
|                                                                   |            | [0, 1]       | [2, 6]    | [2, 11]   | [2, 21]   | [2, 61]   | [0, 61]   |
| Small<br>stocks                                                   | Diff. bet. | Mean         | 0.0216**  | 0.0105**  | 0.0198**  | 0.0272**  | 0.0598**  |
|                                                                   | Q5 and Q1  | $t$ -stat.   | (5.46)    | (2.88)    | (4.13)    | (4.53)    | (5.63)    |
| Mid-cap<br>stocks                                                 | Diff. bet. | Mean         | 0.0152**  | 0.0057*   | 0.0053    | 0.0096*   | 0.0224**  |
|                                                                   | Q5 and Q1  | $t$ -stat.   | (5.21)    | (2.30)    | (1.60)    | (2.15)    | (2.90)    |
| Large<br>stocks                                                   | Diff. bet. | Mean         | 0.0099**  | 0.0074**  | 0.0086**  | 0.0111**  | 0.0197**  |
|                                                                   | Q5 and Q1  | $t$ -stat.   | (3.71)    | (3.76)    | (3.15)    | (2.99)    | (2.95)    |

(a) Table III: Pre-event trading predicts later abnormal returns.

**Table IV**  
**Returns following the Event: Past Returns and Earnings Surprises**

This table presents analysis of market-adjusted returns following earnings announcements conditional on different levels of net individual trading before the event ( $IndNT_{[-10, -1]}$ ) and either past returns (in Panel A) or earnings surprises (in Panel B). We construct the net individual trading measure by first computing an imbalance measure (see Table II) and then subtracting the mean daily imbalance over the sample period. In Panel A, we sort stocks into five quintiles on cumulative market-adjusted return in [-10, -1] ( $CAR_{[-10, -1]}$ ), and within each quintile we sort on net individual trading before the event. We then compute for each stock the cumulative market-adjusted return in [0, 61]. We implement the Fuller-Battese (1974) methodology to correct for clustering: for each of the 25 categories, we model the cumulative abnormal return using a one-way random effect framework in which there is a quarterly effect, and report the estimated means with clustering-corrected  $t$ -statistics (testing the hypothesis of zero cumulative abnormal return). In Panel B, we sort stocks into quintiles on the analyst earnings surprise measure ( $ES$ ), and within each quintile we sort on net individual trading before the event ( $IndNT_{[-10, -1]}$ ). We report for each cell the estimated mean for  $CAR_{[0, 61]}$  with clustering-corrected  $t$ -statistics (in parentheses, from the Fuller-Battese methodology with quarterly clustering). We use \*\* to indicate significance at the 1% level and \* to indicate significance at the 5% level (both against a two-sided alternative).

| Panel A: Cumulative Abnormal Return in [0, 61] Conditional on $CAR_{[-10, -1]}$ and $IndNT_{[-10, -1]}$ |            |                   |           |           |           |                  |                         |
|---------------------------------------------------------------------------------------------------------|------------|-------------------|-----------|-----------|-----------|------------------|-------------------------|
| $IndNT_{[-10, -1]}$                                                                                     |            | $CAR_{[-10, -1]}$ |           |           |           | (Positive)<br>Q5 | Diff. bet.<br>Q5 and Q1 |
|                                                                                                         |            | (Negative)<br>Q1  | Q2        | Q3        | Q4        |                  |                         |
| Q1<br>(Selling)                                                                                         | Mean       | -0.0263*          | -0.0341** | -0.0311** | -0.0348** | -0.0370**        | -0.0107                 |
|                                                                                                         | $t$ -stat. | (-2.09)           | (-4.73)   | (-4.49)   | (-4.78)   | (-3.68)          | (-0.79)                 |
| Q2                                                                                                      | Mean       | -0.0150           | 0.0000    | -0.0195*  | -0.0259** | -0.0322**        | -0.0171                 |
|                                                                                                         | $t$ -stat. | (-1.19)           | (0.00)    | (-2.07)   | (-2.57)   | (-3.42)          | (-1.40)                 |
| Q3                                                                                                      | Mean       | 0.0138            | 0.0137    | -0.0019   | -0.0055   | -0.0052          | -0.0187                 |
|                                                                                                         | $t$ -stat. | (1.34)            | (1.80)    | (-0.18)   | (-0.58)   | (-0.38)          | (-1.40)                 |
| Q4                                                                                                      | Mean       | 0.0339**          | 0.0182    | 0.0185*   | 0.0064    | 0.0197           | -0.0141                 |
|                                                                                                         | $t$ -stat. | (2.95)            | (1.61)    | (2.20)    | (0.77)    | (1.48)           | (-0.95)                 |
| Q5<br>(Buying)                                                                                          | Mean       | 0.0295            | 0.0266*   | 0.0232**  | 0.0026    | 0.0223           | -0.0077                 |
|                                                                                                         | $t$ -stat. | (1.79)            | (2.18)    | (2.97)    | (0.25)    | (1.09)           | (-0.45)                 |
| Diff. bet.                                                                                              | Mean       | 0.0555**          | 0.0611**  | 0.0543**  | 0.0371**  | 0.0584**         |                         |
| Q5 and Q1                                                                                               | $t$ -stat. | (3.37)            | (5.80)    | (5.21)    | (3.49)    | (4.12)           |                         |
| Panel B: Cumulative Abnormal Return in [0, 61] Conditional on $ES$ and $IndNT_{[-10, -1]}$              |            |                   |           |           |           |                  |                         |
| $IndNT_{[-10, -1]}$                                                                                     |            | $ES$              |           |           |           | (Positive)<br>Q5 | Diff. bet.<br>Q5 and Q1 |
|                                                                                                         |            | (Negative)<br>Q1  | Q2        | Q3        | Q4        |                  |                         |
| Q1<br>(Selling)                                                                                         | Mean       | -0.0746**         | -0.0492** | -0.0353** | -0.0192*  | 0.0002           | 0.0758**                |
|                                                                                                         | $t$ -stat. | (-4.54)           | (-5.72)   | (-2.78)   | (-2.46)   | (0.01)           | (5.77)                  |
| Q2                                                                                                      | Mean       | -0.0251           | -0.0277** | -0.0213   | -0.0131   | 0.0133           | 0.0387**                |
|                                                                                                         | $t$ -stat. | (-1.86)           | (-3.29)   | (-1.74)   | (-1.58)   | (1.31)           | (2.95)                  |
| Q3                                                                                                      | Mean       | -0.0146           | -0.0194*  | -0.0051   | 0.0188    | 0.0363**         | 0.0515**                |
|                                                                                                         | $t$ -stat. | (-0.87)           | (-2.30)   | (-0.27)   | (1.95)    | (3.92)           | (3.63)                  |
| Q4                                                                                                      | Mean       | 0.0022            | -0.0053   | 0.0108    | 0.0201*   | 0.0728**         | 0.0715**                |
|                                                                                                         | $t$ -stat. | (0.14)            | (-0.51)   | (1.15)    | (2.03)    | (6.31)           | (4.79)                  |
| Q5                                                                                                      | Mean       | -0.0049           | 0.0075    | 0.0077    | 0.0132    | 0.0732**         | 0.0786**                |
|                                                                                                         | $t$ -stat. | (-0.30)           | (0.72)    | (0.92)    | (1.34)    | (5.74)           | (4.73)                  |
| Diff. bet.                                                                                              | Mean       | 0.0701**          | 0.0596**  | 0.0385**  | 0.0320**  | 0.0730**         |                         |
| Q5 and Q1                                                                                               | $t$ -stat. | (4.41)            | (6.43)    | (3.43)    | (3.18)    | (5.22)           |                         |

(b) Table IV: Predictability remains after conditioning on pre-event returns and earnings surprises.

## 6.3 Table V: Regressions Relating Pre-event Trading of Individuals to Future Abnormal Returns

Read:

- Table V provides regression evidence that pre-event net individual trading predicts  $CAR_{[0, 61]}$ .
- The coefficient on  $IndNT_{[-10, -1]}$  is positive and highly significant.
- Negative earnings surprise indicators predict lower future returns, and positive surprise indicators predict higher returns, but pre-event trading still adds explanatory power.

**Economic message:** Retail order flow is not just correlated with public earnings surprises; it contains additional information about future returns.

**Interpretation:** The regression format shows incremental predictive content in a multivariate setting. This helps rule out the possibility that the sort results are only caused by known earnings-surprise patterns.

## 6.4 Table VI: Decomposition of the Abnormal Returns following Individual Trading

Read:

Table V  
Regressions Relating Pre-event Trading of Individuals to Future Abnormal Returns

Table V presents a regression analysis relating abnormal returns on and after the event to pre-event trading by individuals. The dependent variable in the regressions is the cumulative abnormal return ( $CAR_{[0,61]}$ ), and the regressors include dummy variables for quintiles 1, 2, 4, and 5 earnings surprise measure ( $ES$ ), net individual trading before the event ( $IndNT_{[-10,-1]}$  or  $IndNT_{[-60,-1]}$ ), and past abnormal returns. To control for earnings surprise dummies, we sort stocks into quintiles every quarter on the analyst earnings surprise measure. We construct the net individual trading measure ( $IndNT_{[-10,-1]}$  and  $IndNT_{[-60,-1]}$ ) by first computing a daily imbalance measure (see Table II) and then subtracting an daily imbalance over the sample period; we compute for each stock the cumulative net individual trading measure over these two periods to the announcement. In order to overcome potential econometric problems associated with contemporaneously correlated errors for earnings surprises that are clustered in time, we implement the Fuller-Battese (1974) methodology. This approach uses a one-way random effect model in which there is a quarterly effect, and enables us to compute clustering-corrected  $t$ -statistics (in parentheses) for the coefficients. We use \*\* to indicate significance at the 1% level and \* to indicate significance at the 5% level (both against a two-sided alternative).

| Report | ES1       | ES2       | ES4    | ES5      | IndNT <sub>[-60,-1]</sub> | IndNT <sub>[-10,-1]</sub> | CAR <sub>[-60,-1]</sub> | CAR <sub>[-10,-1]</sub> |
|--------|-----------|-----------|--------|----------|---------------------------|---------------------------|-------------------------|-------------------------|
| 0051   | -0.0217** | -0.0149** | 0.0090 | 0.0444** |                           | 0.0349**                  |                         | -0.0640**               |
| 30     | (-3.64)   | (-2.61)   | (1.51) | (7.46)   |                           | (8.39)                    |                         | (-3.46)                 |
| 0038   | -0.0214** | -0.0152** | 0.0088 | 0.0424** | 0.0107**                  |                           | 0.0077                  |                         |
| 73     | (-3.57)   | (-2.67)   | (1.47) | (7.12)   | (9.88)                    |                           | (0.91)                  |                         |

- Table VI decomposes abnormal returns following individual trading into liquidity-provision and information components.
- The results imply that both channels matter: individual investors earn returns partly because they provide liquidity and partly because their trades contain information.
- The information component is especially strong for small stocks, consistent with the idea that retail investors may have more relative informational advantage where analyst coverage is weaker.

**Economic message:** The table is the bridge between “retail trading predicts returns” and “retail trading is informed.” It shows that the return premium cannot be entirely attributed to liquidity provision.

**Interpretation:** This decomposition is the paper’s most important mechanism test. If all returns were liquidity compensation, the paper would mainly be about retail market making. The residual information component changes the interpretation toward informed or skilled individual trading.

Table VI  
Decomposition of the Abnormal Returns following Individual Trading

This table presents a decomposition of market-adjusted returns following pre-event individual investor trading into a portion attributed to liquidity provision and a portion attributed to information (or skill). For each day  $t$  during the sample period we take all the stocks in our sample that did not have earnings announcements in a 20-day window around that day, and we estimate the following two cross-sectional models:

$$\text{Model 1: } CAR_{t,j+1} = a_1 + b_1 \cdot IndNT_{[-10,-1]} + c_1 \cdot CAR_{[-10,-1]} + \text{error}$$

$$\text{Model 2: } CAR_{t,j+1} = a_2 + b_2 \cdot IndNT_{[-60,-1]} + c_2 \cdot CAR_{[-60,-1]} + \text{error}$$

The models give us estimated parameters that describe the relation between net individual trading and future returns for each day in the sample period. To compute the expected abnormal return due to liquidity provision for a certain earnings announcement, we take the estimated parameters for the day of the announcement from Model 1 and the actual values of net individual trading and returns before the specific earnings announcement and use them to compute the expected abnormal return according to Model 1:

$$ECAR_{t,j+1}^{LP} = a_1 + b_1 \cdot IndNT_{[-10,-1]} + c_1 \cdot CAR_{[-10,-1]}$$

A similar construction produces the estimate  $ECAR_{t,j+1}^{INF}$  using the parameters estimated from Model 2. We follow this process for each earnings announcement in our sample. We also compute for each event a return component that is attributed to information/skill by taking the difference between the actual abnormal return and the estimate of the abnormal return due to liquidity provision ( $CAR_{t,j+1} - ECAR_{t,j+1}^{LP}$  and  $CAR_{t,j+1} - ECAR_{t,j+1}^{INF}$ ). In Panel A, we sort all stocks each quarter into quintiles according to net individual trading in the 10 trading days prior to the announcement ( $IndNT_{[-10,-1]}$ ); quintile 1 contains the stocks that individuals sold the most and quintile 5 contains the stocks individuals bought the most. Since each quarter contains multiple earnings announcements, we implement the Fuller-Battese (1974) methodology to correct for clustering. In Panel B, we separately sort large, mid-cap, and small stocks into quintiles according to net individual trading before the event, and report just the row “Difference between Q5 and Q1” for these size groups. In Panel C, we omit events for which NYSE specialists’ trading activity (either buying or selling) in the 10 days prior to the announcement is high relative to their activity in the previous four 10-day periods (to eliminate events where the price of liquidity provision before the event could have changed significantly). We use \*\* to indicate significance at the 1% level and \* to indicate significance at the 5% level (both against a two-sided alternative). We report clustering-corrected  $t$ -statistics in parentheses.

| Panel A: Return Decomposition into Liquidity Provision and Information Components |                       |                        |                        |             |             |
|-----------------------------------------------------------------------------------|-----------------------|------------------------|------------------------|-------------|-------------|
| IndNT <sub>[-10,-1]</sub>                                                         | CAR <sub>[0,61]</sub> | ECAR <sub>[0,61]</sub> | ECAR <sub>[0,61]</sub> | CAR - ECAR1 | CAR - ECAR2 |
| Q1                                                                                | Mean                  | -0.0318**              | -0.0101**              | -0.0214**   | -0.0226**   |
|                                                                                   | t-stat.               | (-5.14)                | (-2.59)                | (-3.79)     | (-4.16)     |
| Q2                                                                                | Mean                  | -0.0178**              | -0.0002                | -0.0175     | -0.0180     |
|                                                                                   | t-stat.               | (-2.76)                | (-0.06)                | (-1.70)     | (-1.70)     |
| Q3                                                                                | Mean                  | 0.0031                 | 0.0018                 | 0.0021      | 0.0019      |
|                                                                                   | t-stat.               | (0.50)                 | (0.44)                 | (0.25)      | (0.20)      |
| Q4                                                                                | Mean                  | 0.0213**               | 0.0057                 | 0.0156*     | 0.0162*     |
|                                                                                   | t-stat.               | (4.20)                 | (0.91)                 | (2.28)      | (2.01)      |
| Q5                                                                                | Mean                  | 0.0258**               | 0.0171**               | 0.0144*     | 0.0123      |
|                                                                                   | t-stat.               | (3.24)                 | (2.92)                 | (1.98)      | (1.29)      |
| Q5 and Q1                                                                         | Mean                  | 0.0536**               | 0.0278**               | 0.0258**    | 0.0345**    |
|                                                                                   | t-stat.               | (7.83)                 | (23.42)                | (16.91)     | (4.77)      |

Individual Investor Trading and Return Returns

657

Table VI—Continued

| Panel A: Return Decomposition into Liquidity Provision and Information Components            |                     |                       |                        |                        |             |             |          |
|----------------------------------------------------------------------------------------------|---------------------|-----------------------|------------------------|------------------------|-------------|-------------|----------|
| IndNT <sub>[-10,-1]</sub>                                                                    |                     | CAR <sub>[0,61]</sub> | ECAR <sub>[0,61]</sub> | ECAR <sub>[0,61]</sub> | CAR - ECAR1 | CAR - ECAR2 |          |
| Q2                                                                                           | Mean                | -0.0198**             | -0.0010                | -0.0006                | -0.0199*    | -0.0208*    |          |
|                                                                                              | t-stat.             | (-3.50)               | (-0.34)                | (-0.11)                | (-2.98)     | (-2.10)     |          |
| Q3                                                                                           | Mean                | 0.0030                | 0.0017                 | 0.0016                 | 0.0020      | 0.0020      |          |
|                                                                                              | t-stat.             | (0.57)                | (0.44)                 | (0.22)                 | (0.22)      | (0.22)      |          |
| Q4                                                                                           | Mean                | 0.0191**              | 0.0051                 | 0.0045                 | 0.0138*     | 0.0144      |          |
|                                                                                              | t-stat.             | (4.80)                | (1.29)                 | (0.97)                 | (1.97)      | (1.87)      |          |
| Q5                                                                                           | Mean                | 0.0215**              | 0.0174**               | 0.0145*                | 0.0039      | 0.0079      |          |
|                                                                                              | t-stat.             | (2.88)                | (3.07)                 | (2.47)                 | (0.43)      | (0.83)      |          |
| Q5 and Q1                                                                                    | Mean                | 0.0545**              | 0.0286**               | 0.0231**               | 0.0255**    | 0.0323**    |          |
|                                                                                              | t-stat.             | (9.53)                | (29.68)                | (20.46)                | (4.44)      | (5.69)      |          |
| Panel B: Return Decomposition by Market Capitalization Groups                                |                     |                       |                        |                        |             |             |          |
| IndNT <sub>[-10,-1]</sub>                                                                    |                     | CAR <sub>[0,61]</sub> | ECAR <sub>[0,61]</sub> | ECAR <sub>[0,61]</sub> | CAR - ECAR1 | CAR - ECAR2 |          |
| Small stocks                                                                                 | Diff bet. Q5 and Q1 | Mean                  | 0.0603**               | 0.0469**               | 0.0354**    | 0.0344**    | 0.0448** |
|                                                                                              | t-stat.             | (7.01)                | (21.44)                | (14.34)                | (3.92)      | (3.92)      |          |
| Mid-Cap stocks                                                                               | Diff bet. Q5 and Q1 | Mean                  | 0.0331**               | 0.0235**               | 0.0120**    | 0.0116      | 0.0231** |
|                                                                                              | t-stat.             | (4.18)                | (11.28)                | (5.86)                 | (1.37)      | (2.71)      |          |
| Large stocks                                                                                 | Diff bet. Q5 and Q1 | Mean                  | 0.0303**               | 0.0300**               | 0.0199**    | 0.0104*     | 0.0104*  |
|                                                                                              | t-stat.             | (4.13)                | (12.82)                | (6.78)                 | (0.04)      | (2.21)      |          |
| Panel C: Return Decomposition Excluding Announcements with High Specialist Pre-event Trading |                     |                       |                        |                        |             |             |          |
| IndNT <sub>[-10,-1]</sub>                                                                    |                     | CAR <sub>[0,61]</sub> | ECAR <sub>[0,61]</sub> | ECAR <sub>[0,61]</sub> | CAR - ECAR1 | CAR - ECAR2 |          |
| Q1                                                                                           | Mean                | -0.0318**             | -0.0101**              | -0.0094                | -0.0214**   | -0.0226**   |          |
|                                                                                              | t-stat.             | (-5.14)               | (-2.59)                | (-2.39)                | (-3.79)     | (-4.16)     |          |
| Q2                                                                                           | Mean                | -0.0178**             | -0.0002                | -0.0002                | -0.0175     | -0.0180     |          |
|                                                                                              | t-stat.             | (-2.76)               | (-0.06)                | (-0.01)                | (-1.70)     | (-1.70)     |          |
| Q3                                                                                           | Mean                | 0.0031                | 0.0018                 | 0.0021                 | 0.0022      | 0.0019      |          |
|                                                                                              | t-stat.             | (0.50)                | (0.44)                 | (0.45)                 | (0.25)      | (0.20)      |          |
| Q4                                                                                           | Mean                | 0.0213**              | 0.0057                 | 0.0048                 | 0.0156*     | 0.0162*     |          |
|                                                                                              | t-stat.             | (4.20)                | (0.91)                 | (0.71)                 | (2.28)      | (2.01)      |          |
| Q5                                                                                           | Mean                | 0.0258**              | 0.0171**               | 0.0144*                | 0.0088      | 0.0123      |          |
|                                                                                              | t-stat.             | (3.24)                | (2.92)                 | (2.47)                 | (0.86)      | (1.29)      |          |
| Q5 and Q1                                                                                    | Mean                | 0.0570**              | 0.0278**               | 0.0217**               | 0.0296**    | 0.0345**    |          |
|                                                                                              | t-stat.             | (7.83)                | (23.42)                | (16.91)                | (4.66)      | (4.77)      |          |

658

The Journal of Finance

## 6.5 Table VII: Decomposition of the Abnormal Return with Risk Adjustment

Read:

- Table VII repeats the decomposition after risk adjustment.
- The predictive relation remains after accounting for risk characteristics.



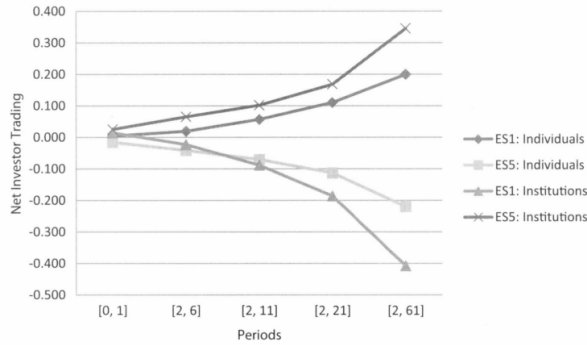
## 6.7 Figure 1: Investor Trading Conditional on Earnings Surprise; Figure 2: Investor Trading Conditional on Earnings Surprise and Pre-event Returns

Read:

- Figure 1 compares individual and institutional trading after positive and negative earnings surprises.
- Individuals tend to buy after bad news and sell after good news; institutions tend to trade in the opposite direction.
- Figure 2 shows that this news-contrarian behavior remains after conditioning on pre-event returns.

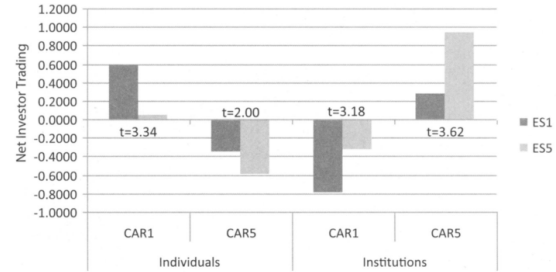
**Economic message:** Individuals behave differently before and after earnings announcements. Before announcements, their trades forecast returns; after announcements, they trade against the public news, potentially slowing price adjustment.

**Interpretation:** The figures are crucial because they separate two mechanisms: pre-event information and post-event contrarian behavior. A retail investor can be informed before the event and still trade against news afterward while unwinding positions or supplying liquidity.



**Figure 1. Investor trading conditional on earnings surprise.** This figure presents analysis of net individual and institutional trading on and after earnings announcements conditional on different levels of the analyst earnings surprise measure. We construct the net individual trading measure by first computing a daily imbalance measure (see Table II) and then subtracting the daily mean imbalance over the sample period. We follow a similar procedure to construct the net institutional trading measure (which excludes dealers and index arbitrageurs). We sort all stocks each quarter into quintiles according to the earnings surprise (*ES1* contains the stocks with the most negative earnings surprise and *ES5* the stocks with the most positive earnings surprise). We then compute for each stock the net investor trading measure for individuals and institutions over certain periods on and after the event. We implement the Fuller–Battese (1974) methodology to correct for clustering: for each quintile, we model the net investor trading measure using a one-way random effect framework in which there is a weekly effect (for [0, 1] and [2, 6]), a monthly effect (for [2, 11] and [2, 21]), or a quarterly effect (for [2, 61] and [0, 61]). We plot the estimated mean for the most extreme quintiles (bad and good news).

(a) Figure 1: Trading conditional on earnings surprise.



**Figure 2. Investor trading conditional on earnings surprise and pre-event returns.** This figure presents analysis of net individual and institutional trading in the post-event period conditional on both different levels of abnormal returns before the announcement and the extent of the earnings surprise. We construct the net individual trading measure by first computing a daily imbalance measure (see Table II) and then subtracting the daily mean imbalance over the sample period. The net institutional trading measure is constructed in an analogous fashion. Earnings surprise (*ES*) is defined as actual earnings minus the earnings forecast one month before the announcement, divided by the price on the forecast day. For the analysis in the figure, we sort stocks independently along two dimensions: market-adjusted returns in the three months prior to the announcement ( $CAR_{[-60, -1]}$ ) and *ES*. We sort the stocks into 25 categories: five groups of earnings surprises and five groups of cumulative abnormal returns. We then compute for each stock the cumulative net individual and institutional trading measures over the period [2, 61]. We implement the Fuller–Battese (1974) methodology to correct for clustering. We then plot the estimated means for the net investor trading measures for the extreme analyst surprise quintiles (*ES1*, bad news, and *ES5*, good news) and the extreme pre-event return quintiles (*CAR1*, most negative, and *CAR5*, most positive). We provide in the figure (next to the columns) the clustering-corrected *t*-statistics from the Fuller–Battese (1974) methodology for the difference between the behavior of investors on *ES1* and *ES5*.

(b) Figure 2: Trading conditional on surprise and pre-event returns.

## 6.8 Table IX: Post-event Individual Trading Conditional on Earnings Surprises and Pre-event Returns; Table X: Regressions Explaining Post-event Net Individual Trading

Read:

- Table IX sorts post-event individual trading by both earnings surprise and pre-event returns.
- It shows that individuals trade in the opposite direction of both prior returns and earnings news.
- Table X confirms this pattern in regressions: post-event individual trading is strongly related to earnings surprises, pre-event individual trading, and past returns.

**Economic message:** Retail investors are not uniformly momentum or attention buyers. After earnings announcements, they appear to trade contrarian to the news, which can contribute to slow price adjustment.

**Interpretation:** These tables are the main evidence for news-contrarian behavior. They also explain why individual trading can have different interpretations before and after the announcement: information acquisition before the event, and contrarian/liquidity behavior after the event.

Table 10

### Post-event Individual Trading Conditional on Earnings Surprises and Pre-event Returns

This table presents estimates of net individual trading in the event period conditional on both different levels of abnormal returns before the announcement and the nature of the earnings surprise. For the analysis in this table, we sort stocks independently along two dimensions: market-adjusted returns in the event window prior to the announcement ( $CAR_{i,t-1}$ ), and the actual earnings surprise ( $ES$ ). We sort by stock size ( $SIZE$ ) comprising the group of earnings surprises and first group of market-adjusted returns. We examine net individual trading over two periods: the event window ( $EW$ ) in Panel A, and the event period ( $EP$ ) in Panel B. We represent the Fisher-Burton (FB) methodology to assess the direction, and report the estimated net with clustering standard errors in parentheses, testing the hypothesis of zero net individual trading. We use \*\*\* to indicate significance at the 1% level and \* to indicate significance at the 10% level (both against a two-sided alternative).

Panel A: Net Individual Trading in  $EW$  by Earnings Surprises and Abnormal Returns in  $EW$  ( $n=146$ )

| $CAR_{i,t-1}$ | Earnings Surprises | (Positive) | Diff. bet. Q1 and Q2 |          |          |           |          |          |          |
|---------------|--------------------|------------|----------------------|----------|----------|-----------|----------|----------|----------|
|               | Q1                 | Q2         | Q3                   | Q4       | Q5       | Q6        | Q7       | Q8       | Q9       |
| Q1            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q2            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q3            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q4            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q5            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q6            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q7            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q8            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q9            | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q10           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q11           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q12           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q13           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q14           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q15           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q16           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q17           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q18           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q19           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q20           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q21           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q22           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q23           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q24           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q25           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q26           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q27           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q28           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q29           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q30           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q31           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q32           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q33           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q34           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q35           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q36           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q37           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q38           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q39           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q40           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q41           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q42           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q43           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q44           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q45           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q46           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q47           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q48           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q49           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q50           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q51           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | 0.0001**  | 0.0001** | 0.0001** | 0.0001** |
| Q52           | Mean               | 0.0001**   | 0.0001**             | 0.0001** | 0.0001** | -0.0001** | 0.0001** | 0.0001** | 0.0001** |
|               | Stdev              | 0.0001     |                      |          |          |           |          |          |          |

noisy, the aggregate order imbalance can be informative. This information is most valuable around earnings announcements, when firm fundamentals are about to be revealed.

### **8.3 Why the distinction between pre-event and post-event behavior matters**

The paper’s strongest insight is that retail trading has different meanings at different times. Before earnings announcements, individual buying predicts good news or positive future returns. After announcements, individual investors trade against earnings news and prior returns. The same investor class therefore appears informed in one window and contrarian in another. This distinction helps reconcile informed-retail evidence with behavioral evidence on retail trading.

### **8.4 Implications for market efficiency**

If individuals sell after good earnings news and buy after bad news, their post-event trading can delay the full incorporation of information into prices. This provides a potential microstructure channel for post-earnings-announcement drift. At the same time, pre-event individual trading helps reveal information before public announcements. The paper therefore suggests that individual investors can both improve and slow price discovery depending on timing.

### **8.5 Limitations and caveats**

The evidence is predictive rather than experimental. The decomposition relies on structural assumptions about liquidity provision and return generation. The sample covers NYSE stocks from 2000–2003, and trading technology, market structure, retail participation, and algorithmic intermediation have changed since then. The paper also studies aggregate individual order flow, so it cannot identify which individual investors are informed.

### **8.6 Takeaway**

The lasting contribution is a more nuanced view of individual investors. Retail trading is neither purely noise nor uniformly informed. Aggregated retail order flow before earnings announcements contains economically meaningful information, while post-announcement retail trading is contrarian and may slow price adjustment. This makes the paper important for understanding information production, liquidity provision, and the role of individual investors in price discovery.